



Combustion and Flame

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OUTLINE

- ① [Introduction to Combustion](#)
- ② [Classification of Combustion](#)
- ③ [Conditions for Combustion](#)
- ④ [Ignition Temperature](#)
- ⑤ [Combustible and Non-Combustible Substances](#)
- ⑥ [Fuel](#)
- ⑦ [Flame](#)
- ⑧ [Cooling and Heating](#)
- ⑨ [Fire Control and Extinguishing](#)
- ⑩ [Summary](#)

WHAT IS COMBUSTION?

Definition

Combustion is the process of rapid burning of a substance. It is an exothermic process, i.e., heat is produced during combustion.

- Combustion involves chemical reaction between fuel and oxygen
- Produces heat and light as byproducts
- Can be slow (like rusting) or fast (like burning)

Key Point

The process in which a substance combines chemically with oxygen or any other supporter of combustion, with simultaneous evolution of heat and light is called combustion.

TYPES OF COMBUSTION

Based on Speed:

- Rapid Combustion
- Spontaneous Combustion
- Complete Combustion
- Incomplete Combustion

Based on Oxygen Supply:

- Complete Combustion
- Incomplete Combustion

Rapid Combustion

Definition

The process in which a substance combines chemically with oxygen at a temperature above its ignition temperature and a large amount of heat and light is released.

Examples:

- Burning of wood
- Burning of coal
- Burning of LPG (contains mainly butane)

Reaction



Spontaneous Combustion

Definition

A combustion reaction in which no light is produced and temperature of the substance remains almost unchanged, and slow or spontaneous combustion occurs.

Characteristics:

- No external heat source required
- Substance undergoes slow combustion without catching fire
- Examples: Rusting of iron, digestion of food

Important

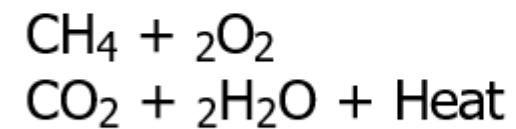
Oxidation at room temperature is called spontaneous combustion.

Complete vs incomplete combustion

Complete Combustion

- Sufficient oxygen available
- Complete burning occurs
- Produces CO₂ and H₂O
- More heat produced
- Clean burning

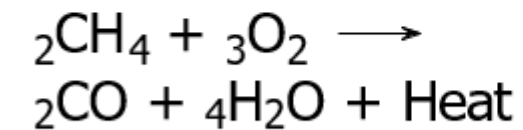
Example



Incomplete Combustion

- Insufficient oxygen supply
- Partial burning occurs
- Produces CO and carbon particles
- Less heat produced
- Sooty flame

Example



Requirements for combustion

Three Essential Requirements

For combustion to occur, three conditions must be met simultaneously:

① Presence of Combustible Substance

- The substance that can burn (fuel)
- Examples: Wood, coal, petrol, kerosene, LPG

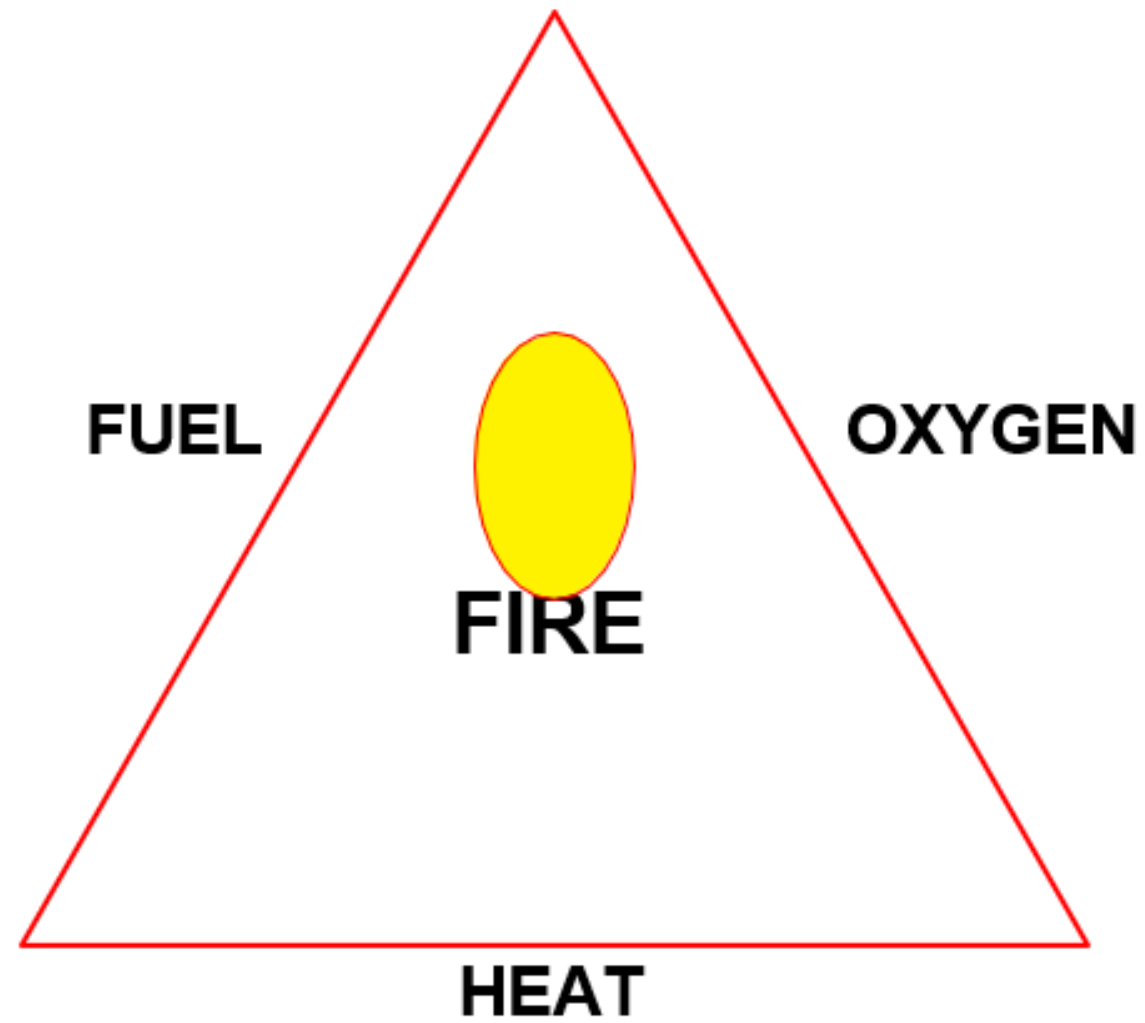
② Presence of Supporter of Combustion

- Usually oxygen from air
- In most cases, oxygen is available from air

③ Heating to Ignition Temperature

- Minimum temperature required to start burning
- Also called ignition point or kindling temperature

Fire Triangle



Key Concept

Remove any one of these three elements and combustion will stop!

Ignition temp

Definition

The minimum temperature at which a substance catches fire and starts burning is known as its ignition temperature or ignition point or kindling temperature.

Key Points:

- Different substances have different ignition temperatures
- A substance must be heated to its ignition temperature to start burning
- Below ignition temperature, combustion cannot occur

Examples

Paper: 230°C

Wood: 300°C

Coal: 500°C

Kerosene: 210°C

Classification of Substances

Combustible Substances

- Can burn readily in air
- Examples:
 - Petrol, LPG, cooking gas
 - Wax, kerosene, paper
 - Cloth, wood, coal

Non-Combustible Substances

- Cannot burn in air
- Examples:
 - Water, glass, sand
 - Stone, iron

Note

The substances which do not burn are called non-combustible substances.

What is fuel?

Definition

A combustible substance which on burning produces a large amount of heat and light is called a fuel.

Examples: Coal, LPG, Petrol, Kerosene, Wood, etc.

Classification of Fuels

Fuels are classified on the basis of physical state in which they occur. So fuels are classified as solid, liquid and gaseous fuels.

Types of fuel

Solid Fuels

- Coal
- Wood
- Charcoal
- Dung cakes

Liquid Fuels

- Petrol
- Diesel
- Kerosene
- Alcohol

Gaseous Fuels

- Natural gas
- LPG (Liquefied Petroleum Gas)
- Coal gas
- Hydrogen gas

Advantage of Gaseous Fuels

Gaseous fuels are considered the cleanest burning fuels with highest efficiency.

Ideal fuel

An ideal fuel should have the following characteristics:

- It should be fairly cheap and easily available
- It should burn at moderate rate
- It should not produce any poisonous and irritating fumes during burning
- It should leave no residue (ash) after burning
- It should produce large amount of heat per unit mass
- It should be safe and convenient from the storage and transportation point of view
- Its ignition temperature should be above room temperature

Note

No single fuel satisfies all these conditions perfectly.

Introduction to flame

Definition

When you light a matchstick, it burns with a yellow flame. The flame produced by burning LPG is blue.

Key Points:

- Flame is produced by burning gases
- Different gases produce different colored flames
- Solid combustible substances burn with a flame only when heated to produce vapors
- Liquid combustible fuels also burn with a flame

Examples

Kerosene burns with a flame

Coal burns with a flame (after producing vapors)

Charcoal burns without a flame (glows)

Types of flames

Luminous Flame

- Produces bright yellow light
- Incomplete combustion occurs
- Less hot than non-luminous flame
- Produces soot and carbon particles

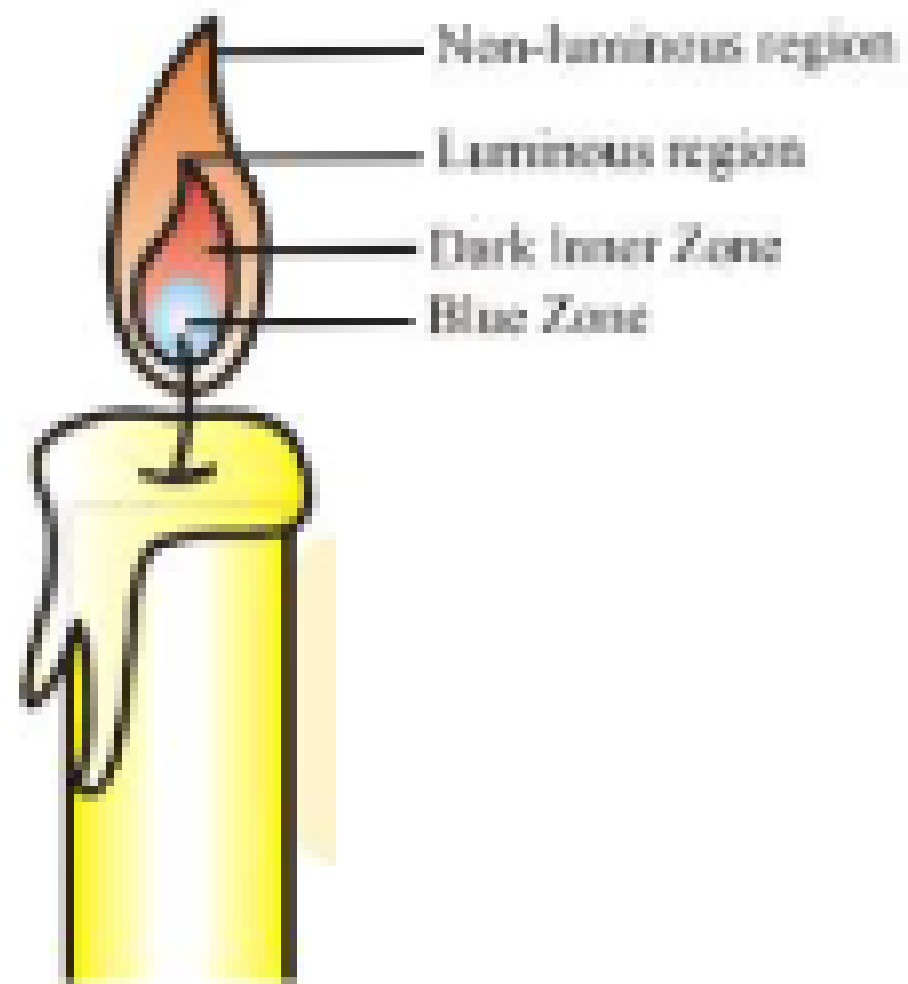
Non-Luminous Flame

- Produces little or no light
- Complete combustion occurs
- Very hot flame
- Clean burning, no soot production

Comparison

Non-luminous flames are preferred for heating and cooking as they produce more heat and are cleaner.

Structure of candle flame



Flame of a candle

Zones

① Innermost Zone (Dark Zone)

- Located at the base of the flame
- Least hot zone
- Contains unburnt wax vapors
- No combustion takes place

② Middle Zone (Luminous Zone)

- Bright yellow colored zone
- Incomplete combustion occurs
- Contains partially burnt carbon particles
- Moderately hot

③ Outermost Zone (Non-luminous Zone)

- Hottest zone of the flame
- Complete combustion takes place
- Blue in color
- Clean burning

Cooling and Heating

Common Uses of Fire

The most common use of fire is for cooking and heating. Fire is commonly used for:

- **Cooking:** Boiling, frying, baking, roasting
- **Heating:** Warming homes, heating water
- **Industrial purposes:** Melting metals, manufacturing
- **Transportation:** Fuel for vehicles, aircraft
- **Electricity generation:** Thermal power plants

Safety Note

Fire should always be used with proper safety measures and adult supervision.

Methods of fire extinguishing

Principle

To extinguish fire, we need to remove one or more elements of the fire triangle.

① Removal of Fuel

- Turn off gas supply
- Remove combustible materials

② Removal of Oxygen

- Cover with sand, foam, or CO₂
- Use fire blankets

③ Cooling Below Ignition Temperature

- Spray water
- Use cooling agents

Types of fire extinguishers

- **Water:** Most common, works by cooling
 - Never use on electrical fires
 - Never use on oil fires
- **Foam:** Forms a blanket over burning liquid
- **Carbon Dioxide (CO₂):** Displaces oxygen
 - Good for electrical fires
 - Leaves no residue
- **Dry Chemical Powder:** Interrupts chemical reaction
- **Sand:** Cuts off oxygen supply, absorbs heat

Key points

- Combustion is a chemical process that produces heat and light
- Three conditions are essential: fuel, oxygen, and ignition temperature
- Fuels can be solid, liquid, or gaseous
- Complete combustion is more efficient than incomplete combustion
- Flames have different zones with different temperatures
- Fire can be controlled by removing fuel, oxygen, or heat
- Safety measures are essential when dealing with fire

Remember

Fire is a good servant but a bad master. Always use fire safely!

Thank You!

Questions?